

The Puzzle of Language Acquisition

(Caroline Rowland)

17/08/20

Neural vs Computational Cognition

How to simulate language acquisition processes in connectionism models??

① Setting the empirical landscape

Animals with
LANGUAGE

- Humans

Animals without
LANGUAGE

- The rest

~ 3.000.000.000 years to learn
as GPT5

What do need to learn?

- ☐ Produce phonemes
- ☐ Learn words
- ☐ Learn grammar
- ☐ Learn how to communicate

The word-referent mapping problem



glorp

Half of the time ~~see~~ child look at the correct "GLORP" (interpret the speaker behaviour)

→ The extension problem

Glorp = a cat
≠ a dog



[Cross linguistic differences in the extension problem]

③ Learn grammar

The dog bit the man

Der Hund beißt den Mann

Den Mann beißt der Hund

④ Learn how to communicate

I Feel like pizza → like "having pizza"



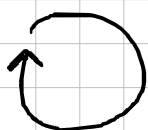
Like lack some common sense

Possible solutions that don't work

- 1) PreFormism  → HOMUNCULUS THAT KNOWS LANGUAGE
- 2) Simple Imitation (1957) → We GENERATE
"Colourless green ideas sleep furiously"
- 3) Explicit Teaching
• CAVEAT: Recant seem to work
- 4) Instantaneous Parameter setting

Possible solutions that may work

1) A Framework



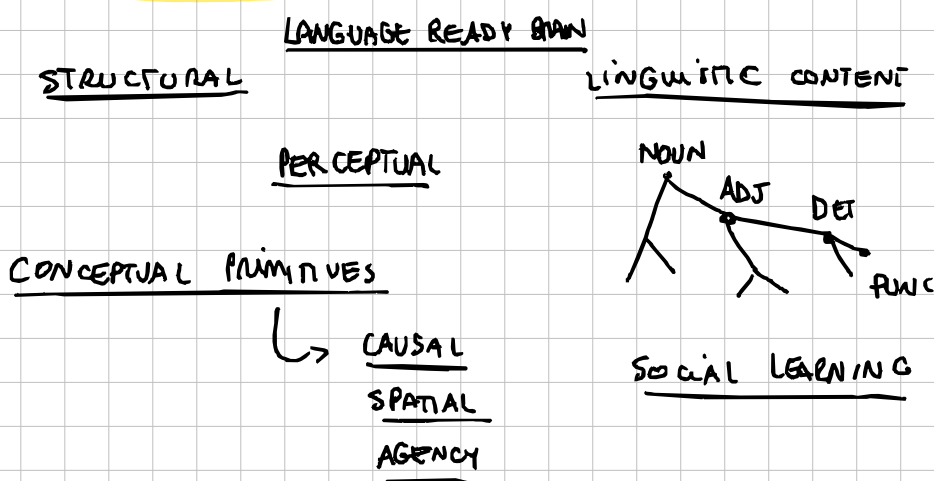
- MULTIMODAL INPUT
- STRUCTURE BUILDING
- ACTIVE, ADAPTIVE LEARNING
- DYNAMIC, DEVELOPMENTAL CHANGE

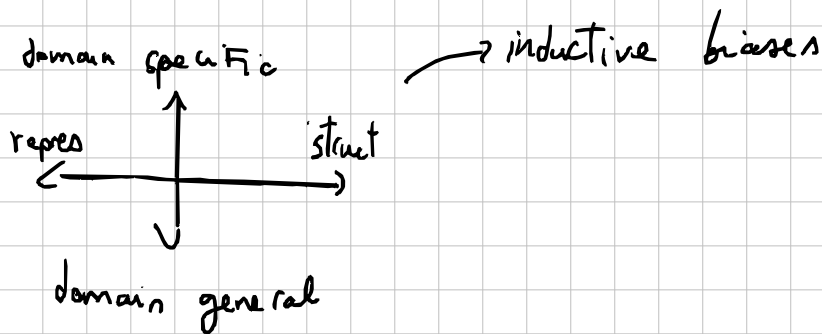
Innatism?

Chomsky vs Skinner

Tommasello vs Pinker

... more like CONSTRAINTS!





18/08/2027

The structure in childrens inputs

Multimodal

→ INPUT Frequency is Key to building linguistic structure

5 Frequency effects

1. Level and Kinds

→ Frequent terms will be learned first
→ Concreteness

2. Age of Acquisition

Wordbank
→ [Age of Acquisition]

3. Prevent error

4. Cause error

5. Interact with other effect

Frequency ↑
Babiness ↑
Concreteness ↑

Mult: model input $\rightarrow$ structure building



optimal input changes as we learn from the input

18 months $\rightarrow$ quantity
30 months $\rightarrow$ diversity
42 months $\rightarrow$ decontextualise talk

CLASSIC (Chunking lexical and sublexical sequences in children)

Real mother input

Quantity manipulated input

Diversity manipulated input

The more you know, the faster you learn

Role of multimodal interaction

Children learn language in dynamic social interactions

Gesture + speech →



Gestures reduce referential ambiguity during naming events



At least if the kid has a small vocabulary

Modeled as a Bayesian process

"Infinite use of finite means"

- Huge cross-linguistic differences across languages in grammar
- Substantial cross-linguistic differences in acquisition timeline

20/08/26

The emergence/construction of grammar

→ What needs to be explained? Many!

- 1) Wh-question errors
- 2) Novel verb studies

Modern Theoretical divisions

Formalist

Grammar is an autonomous formal system independent of meaning.

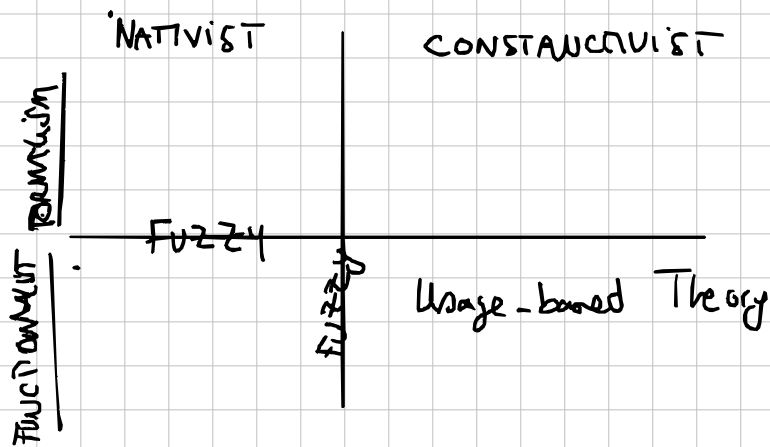
Functionalist

Grammar is shaped by an explicable in terms of the communicative function it serves.

Nativist Innate grammatical knowledge

Constructivism

Grammatical knowledge is built over time



The Language Acquisition device → species-specific

↓
POVERTY OF THE
Stimulus

- 1) Innate UNIVERSAL PRINCIPLES
- 2) Map innate knowledge to the language you are hearing

The Variational learning Model (Yang, 2002)

↳ Parameters associated with PROBABILITIES

Evidence from novel verb studies

The RABBIT is Krobbing The DUCK

- Children tend to not generalize to a new verb Form → no PRODUCTION
- They comprehend!

Many path lead to individual differences but some language

→ Conductive Vocabulary

↓
WHAT FACTORS PREDICT this?

- Many differences even if kids ~~where~~ have very similar history.
- What explains variations :

→ correlates with LANGUAGE knowledge

[INPUT QUANTITY
QUALITY]

• INTERACTION IN CONVERSATIONS

[TURN TAKING]

• Socio-COGNITIVE SYSTEMS

[Lexical processing speed]

• Genes via gene expression

Phonological working memory capacity

- Briefly holds and processes speech sound and verbal info
- Limited capacity
- Capacity is measured in non-word repetition

Can we simulate, in a computational model?

CHUNKING Theory

"CLASSIC"

Input: Phonemic version

101 11 41 81 921 11

101 41 832 1 1

} chunk
and remember
easily

↳ useful for children!

21/08/2026

- Are LLM learning like modes
- Same Representation?

What kind of input should we feed our models?
[Frequency effects are everywhere]

- Distributional cue may facilitate learning
- Multimodal input? (even touch apparently)
- Social-cultural environment (communities)
- Innate constraints
- Pattern can be already seen in the Brain even if kids cannot speak
- Fleeting memory Transformer, recent post perfectly
- Curiosity driven Reinforcement
- Learning is shaped by development
- Couldn't model all of them

